

saving work 88
 and Scratch 87, 101, 102–5,
 124, 125
 scripts 101
 shell and code windows 92,
 93, 106–7
 shortcuts 171
 Silly sentences project 132–3
 simple commands 102–3
 sorting in 151
 strings in 110, 114–15, 117
 tuples and dictionaries 134–5
 turtle graphics 87
 types of data 110–11
 Unicode in 185
 variables 99, 101, 108–9, 116
 variables and functions 138–9
 website 89
 what is Python? 19, 86–7
 what next? 176–7
 while loops 124–5

Q

questions
 asking 54
 combining 63
 true or false 62–3
 quote marks
 errors 94
 in lists 128
 in strings 110, 114, 116

R

radio waves 195
 RAM 189
 “randint()” function 98, 99, 101,
 104, 113, 130, 153, 155
 Random Access Memory
 see RAM
 Random module 152, 153, 157
 random numbers 53, 104, 113,
 152, 155
 random positions 43, 57
 Raspberry Pi 25, **214**
 reacting to events 162–3
 readable code 133
 remainders 53
 repeat loops 46, 122, 211
 “repeat until” block 68, 76, 77,
 78, 124
 rewriting code 217
 robots 16, 203, 217

home-built 215
 Roll the dice project 60–1
 rotation
 styles 38
 tools 61
 routers 194, 195
 Ruby 18
 running
 programs 23, 102, 106
 scripts 30
 runtime errors 148

S

saving your work 11
 in Python 88, 93, 106, 107
 in Scratch 24, 25, 33
 saying things (in Python) 161
 score 50, 79, 99, 172–3
 scoreboards 172
 Scratch **20–83**
 account 24
 adding sounds **58–9**, 79
 backpack 82
 blocks 30–1, 72–3
 coordinates 56–7
 complex loops 68–9
 costumes 40–1
 creating blocks 72–3
 decisions and branches 64–5
 Escape the dragon! project
 32–7
 events 44–5
 experimenting with 82–3
 harder commands 104–5
 help menu 83
 hide and seek 42–3
 installing and launching 24–5
 interface **26–7**, 49
 lists 55
 make some noise 58–9
 making things move 38–9
 math in 52–3
 menu and tools 26
 and microphones 45
 Monkey mayhem project
 74–81
 pens and turtles **48–9**, 87
 and Python 87, 102–5, 124, 125
 and Python’s Ghost game 101
 Roll the dice project 60–1
 scripts and colored blocks
 30–1, 101
 sending messages 70–1
 sensing and detecting 66–7

simple commands 102–3
 simple loops 46–7
 software 24
 special effects 42–3
 sprites 28–9
 strings and lists 54–5
 true or false 62–3
 typical programs 23
 understanding 22
 variables **50–1**, 108
 versions of 25
 and webcams 45
 website 24, 82
 what is Scratch? 18, **22–3**

screens 181
 scripts 22, 23
 building 27
 and colored blocks 30–1
 flow of 30
 pausing 69
 in Python 101
 repeating 68
 running 30
 in Scratch **30–1**, 101
 and sprites 28
 stopping 30, 68
 testing 30
 scripts area 27
 scripts tab 27
 scriptwriters 204
 SD cards 215
 secret data 202
 selection sort 151
 “Sensing” blocks 31, 34, 36, 51,
 65, 66–7, 75
 separators 117
 servers 219
 shapes
 adding color 159
 drawing 140
 making 158–9
 moving 160
 naming 160
 shell windows 92, 93, **106–7**, 116
 errors in 95
 silicon chips 188, 189
 Silly sentences project 132–3
 skipping, in loops 127
 slicing 115
 smartphones 206
 social apps 206
 Socket module 152
 sockets **195**, 219
 software 14, 15, 219
 sorting algorithms 150–1
 “Sound” blocks 31, 58–9

sound designers 204
 sound library 79
 sounds 22, 23
 adding to program **58–9**, 79
 detectors 45
 in games 205
 playing 58
 selecting from library 37, 58
 volume control 58
 spam 213
 speakers 181
 speech bubbles 22, 28, 87, 101
 adding 41, 161
 conversations 71
 speed, setting 51, 77
 “split()” function 144
 sprites 22, 23, **28–9**, **39**, 219
 adding graphic effects 43
 adding more 76–7, 80
 adding sounds 58–9
 changing size 43
 changing time or speed 35, 51
 coordinates 56–7
 collisions 67
 communication between
 70–1
 controlling 31, 66
 copying or deleting 29
 creating and editing **29**, 34, 36
 designing your own 39
 direction 39
 and events 44–5
 hiding and showing 42
 in the interface 28
 movement 22, 23, **38–9**, 57,
 66–7
 naming 29
 pen tools 48–9
 renaming 29
 rotation styles 38
 and scripts 28
 selecting from library 34, 36,
 39, 75, 76
 and “Sensing” blocks 66–7
 sprite list 27
 turtle graphics 49
 and variables 35, **50–1**
 what they can do 28
 square roots 53
 stage 23, 25, 27
 Standard Library modules 152
 stop button 30
 stopping programs 102
 strategy games 205
 strings 219
 adding 114